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The goal of this exercise sheet is to illustrate and understand **risk decomposition** and the **Bayes predictor**. We denote with $\mathcal{X} = \mathbb{R}$ and $\mathcal{Y} = \mathbb{R}$ the input and output spaces, respectively, and are interested in finding a prediction function $f : \mathcal{X} \longrightarrow \mathcal{Y}$ from a predefined model class $\mathcal{F} = \{g : \mathcal{X} \longrightarrow \mathcal{Y} \text{ subject to some constraints}\}$. Furthermore, let (X, Y) denote the random vector of input-output pairs with common probability distribution $p_{(X,Y)}$ on $\mathcal{X} \times \mathcal{Y}$. We are interested in minimising the *expected risk* $\mathcal{R}(f) = \mathbb{E}_{(X,Y)}[\ell(Y, f(X))]$ with the least-squares loss $\ell(y, z) = (y-z)^2$. Recall that we have the error decomposition

$$\begin{aligned}\mathcal{R}(\hat{f}) - \mathcal{R}^* &= \Big(\mathcal{R}(\hat{f}) - \inf_{f' \in \mathcal{F}} \mathcal{R}(f')\Big) + \Big(\inf_{f' \in \mathcal{F}} \mathcal{R}(f') - \mathcal{R}^*\Big) \\ &= \text{estimation error} + \text{approximation error},\end{aligned}$$

where $\mathcal{R}^* = \mathcal{R}(f_*)$ denotes the Bayes risk (see Task 1). We make the following assumptions on our data (note that these are not known in practice):

$$Y \mid X = x \sim \mathcal{N}(x^2, \sigma^2), \quad X \sim \mathcal{N}(0, \rho^2).$$

Task 1 (Bayes predictor)

1. Write down $\mathbb{E}_Y(Y \mid X = x)$.
2. Compute, conditioned on $X = x$, the Bayes predictor:
$$f_*(x) \in \arg\min_{z \in \mathbb{R}} \mathbb{E}_Y\big[\ell(Y, z) \mid X = x\big].$$

Task 2 (Approximation error)

We consider the model class of polynomials up to degree $k \in \mathbb{N}_0$, i.e.

$$f_\theta(x) = \theta_0 x^0 + \theta_1 x^1 + \cdots + \theta_k x^k.$$

Compute

$$\inf_{\theta \in \mathbb{R}^{k+1}} \mathcal{R}(f_\theta),$$

for $k = 0, 1, 2$. **Bonus:** What happens for $k > 2$?

Hint:

1. Compute $\mathbb{E}_Y[\ell(Y, f(x)) \mid X = x]$.
2. Use the law of total expectation to compute $\mathbb{E}_{(X,Y)}[\ell(Y, f_\theta(X))]$.
3. Minimise over θ.

Task 3 (Estimation error)

We want to investigate the model from Task 2 numerically.

1. Create a dataset of 20 realisations of (X, Y). Divide it into training and testing datasets of equal sizes.
2. Use the polynomial model from Task 2 with $k = 0, \ldots, 10$ on the training dataset to obtain a least-squares fit. Hint: use Python together with scikit-learn. In particular, the functions "PolynomialFeatures" and "linear_model.LinearRegression()".
3. Compute the empirical risk $\hat{\mathcal{R}}(\hat{f}_\theta) = \frac{1}{n}\sum_{i=1}^n \ell(y_i, \hat{f}_\theta(x_i))$ on the training and testing sets.